MAMUNA CHAUDHRY

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EDUCATION

The City College of New York New York, NY

Bachelor of Science in Computer Science Expected December 2027

Relevant Courses: Data Structures, Algorithms, Software Design Lab, Website Design, Probability & Statistics, Data Science

SKILLS

Languages: Python•Javascript•SQL•Typescript•Java•C++•HTML•CSS

Frameworks: Flask•React•NodeJS•Pandas•NumPy•Matplotlib•Seaborn•NetworkX•Material UI

Tools: AWS•Figma•scikit-learn•Unix•Firebase•MySQL•MongoDB•Vercel•Git•Jupyter•Google Analytics•Jira•Agile

Certifications: ML & Data Science (Unsupervised,Time Series Data,Supervised),DSA,Customer Analytics & A/B Testing (2025)

EXPERIENCE

The City College of New York | *Software Engineer Research Fellow* | *New York, NY* January 2025 - September 2025

- Optimized assistive application by integrating a new Text-to-Speech API, reducing response time by **50%**
- Conducted **customer discovery, user research, & testing** at vision rehab centers to **optimize feature** accessibility
- Applied **prompt** engineering for large language models (LLMs) to enhance accuracy and contextual relevance
- Analyzed **quantitative** and **qualitative** data to develop an assistive tech research **paper** on **key findings**

National Intelligence University | *Data Science Research Intern* | *Remote* January 2024 - May 2024

- Built a data tool using CAMEO taxonomy & GDELT data to analyze global event data and **support** research
- Preprocessed **600K**+ records and refactored backend, cutting response time by **20%** and code length by **30%**
- Implemented a **network** graph algorithm to find direct/indirect connections between entities
- Integrated article frequency counts and source URLs for each entity pair to **enhance analytical** insight
- Presented project scope to **data scientists** in the intelligence community and findings in weekly Agile **briefings**

neARabl Inc. | *Data Science Intern* | *New York, NY* July 2023 - August 2023

- Developed an **ETL** pipeline to process **800K**+ socioeconomic data records for a construction application
- Engineered features and optimized data workflows to support **ML** model integration
- Built **35** visuals using Python to **reveal patterns** and improve decision-making through data-driven storytelling

PROJECTS

Buddy Walk | *MongoDB, React, NodeJS, Typescript, Docker, Vite, Figma, GCP* | GitHub | Live App Sep 2024 - Present

- Deployed a **multimodal** assistive app to help Blind and Low Vision (BLV) users travel independently
- Integrated GPT-4 via OpenAI API with Google Maps API to deliver real-time geolocation-based assistance
- Developed an **accessible** user interface with **mockups** using Figma to **ensure usability** for visually impaired users

PropertyIntel | *Flask, React, Supabase, Figma, Vercel, cURL* | GitHub | Live App Oct 2024 - Dec 2024

- Deployed a real estate app with 2 engineers to find real-time NYC building complaints & violations
- Developed a RESTful API with **12** endpoints and **3** search filters for retrieving **2.9M**+ records, using cURL for testing
- Implemented geocoding & Google Maps API for a map-based interface, **boosting** user engagement by **20%**

PUBLICATIONS & AWARDS

- A Multimodal AI-Powered Travel Assistant Geo-App for BLV Users - Accepted to Journal Track, 41st Annual California State University, Northridge (CSUN) Assistive Technology Conference (2026). Recipient of the **Dr. Arthur I. Karshmer Award for Assistive Technology Research** (**Best Paper**)
- **CUNY Student Inclusion Initiatives Award** - Supporting Assistive Technology Research and Education (2025)
- **1st Place** – BMCC Open Data Science Hackathon (2023)